

GROUP CONTRIBUTION SHEET

Gym Membership Management System - CCS2300 Data Structures and Algorithms

Group Number: 17

Submission Date: 08/16/2026

Member 1

Member	Hewage Maleesha Hirumal
Student ID	CIT-25-01-0337
Primary Component	Arrays, Singly Linked List, Linear Search
File(s)	Member1_ArraysLinkedList.java
Contribution Summary	Implemented the fixed Array of membership plans and the dynamic Linked List used to store all registered members. Implemented insert, delete, traverse, and Linear Search methods. Wrote the complexity analysis for this component. Contributed to system testing, integration, and documentation. DatabaseConnection.java (MySQL connection, load/save logic) gym_db.sql (database schema)
Estimated Contribution	40%

Member 2

Member	Matharage don Buddhima Deneth
Student ID	CIT-25-01-0574
Primary Component	Stack, Queue, Bubble Sort
File(s)	Member2_StackQueue.java
Contribution Summary	Implemented the array-based Stack (undo log) and circular array-based Queue (Trainer Waiting Queue). Implemented Bubble Sort. Wrote the complexity analysis for this component. Contributed to system testing, integration, and documentation. All the diagram Drawing (UML class diagram, ERD , Flow chart)
Estimated Contribution	20%

Member 3

Member	Duwage Ashitha Lakshan Chathuranga Perera
Student ID	CIT-25-01-0348
Primary Component	Binary Search Tree (BST), Selection Sort
File(s)	Member3_BST.java
Contribution Summary	Implemented the Binary Search Tree, including insert, search, delete, and inorder/preorder/postorder traversals. Implemented Selection Sort. Wrote the complexity analysis for this component. Contributed to system testing, integration, and documentation.
Estimated Contribution	10%

Member 4

Member	Madduma wellalage Rachintha rashmika
Student ID	CIT-25-01-0356
Primary Component	AVL Tree, Insertion Sort
File(s)	Member4_AVLTree.java
Contribution Summary	Implemented the self-balancing AVL Tree, including all four rotation cases (Left-Left, Right-Right, Left-Right, Right-Left). Implemented Insertion Sort. Wrote the complexity analysis for this component. Contributed to system testing, integration, and documentation.
Estimated Contribution	10%

Member 5

Member	Patti Arachchige Aveesha vimod
Student ID	CIT-25-01-0569
Primary Component	Graph (Referral Network), BFS, DFS, Merge Sort
File(s)	Member5_Graph.java
Contribution Summary	Implemented the referral network as a directed adjacency-list graph, including Breadth-First Search and Depth-First Search. Implemented Merge Sort. Wrote the complexity analysis for this component. Contributed to system testing, integration, and documentation.
Estimated Contribution	10%

Member 6

Member	Horagalage Savindu sangith lakshan perera
Student ID	CIT-25-01-0345
Primary Component	Hash Table, Set ADT, Quick Sort
File(s)	Member6_HashSet.java
Contribution Summary	Implemented the Hash Table with separate chaining for collision handling, and the Set ADT for duplicate email prevention. Implemented Quick Sort. Wrote the complexity analysis for this component. Contributed to system testing, integration, and documentation.
Estimated Contribution	10%

Shared Contributions (All Members)

- System design and architecture discussions
- Main.java (CLI menu, login screen, integration of all components)
- Testing across all menu options and edge cases
- Project Progress Video
- Final Project Report
- Practical demonstration preparation
- All the diagram design (UML class diagram, ERD , Flow chart)